



Title

lprobust — Local Polynomial Regression Estimation with Robust Bias-Corrected Confidence Intervals and Inference Procedures.

Syntax

```
lprobust yvar xvar [if] [in] [, eval(gridvar) neval(#) deriv(#) p(#) h(hvar)
b(bvar) rho(#) kernel(kernelfn) bwselect(bwmethod) bwcheck(#) imsegrid(#)
vce(vcetype [lvceopt]) level(#) bwregul(#) separator(#) interior genvars
covgrid plot graph_options(gphopts) ]
```

Description

lprobust implements local polynomial regression point estimators with robust bias-corrected confidence intervals and inference procedures developed in [Calonico, Cattaneo and Farrell \(2018\)](#). See also [Calonico, Cattaneo and Farrell \(2022\)](#) for related optimality results. It also implements other estimation and inference procedures available in the literature. See Wand and Jones (1995) and Fan and Gijbels (1996) for background references.

A detailed introduction to this command is given in [Calonico, Cattaneo and Farrell \(2019\)](#).

Companion command is: [lpbwselect](#) for data-driven bandwidth selection.

Related Stata and R packages useful for empirical analysis are described in the following website:

<https://nppackages.github.io/>

Options

eval(gridvar) specifies the grid of evaluation points for xvar. By default it uses 30 equally spaced points over to support of xvar.

neval(#) specifies the number of evaluation points to estimate the regression functions. Default is 30 evaluation points.

deriv(#) specifies the order of the derivative of the regression functions to be estimated. Default is **deriv**(0).

p(#) specifies the order of the local polynomial used to construct the point estimator. Default is **p**(1) (local linear regression).

h(hvar) specifies the main bandwidth (h) used to construct the point estimator for each evaluation point. If not specified, it is computed by the companion command [lpbwselect](#).

b(bvar) specifies the bias bandwidth (b) used to construct the bias-correction estimator for each evaluation point. If not specified, it is computed by the companion command [lpbwselect](#).

rho(#) specifies the value of ρ , so that the bias bandwidth b equals $b=h/\rho$. Default is **rho**(1) if h is specified but b is not.

kernel(kernelfn) specifies the kernel function used to construct the local-polynomial estimator(s). Options are: **triangular**, **epanechnikov**, **uniform** and **gaussian**. Default is **kernel**(epanechnikov).

bwselect (*bwmethod*) bandwidth selection procedure to be used. By default it computes both *h* and *b*, unless *rho* is specified, in which case it only computes *h* and sets *b*=*h*/*rho*. Options are:

- mse-dpi** second-generation DPI implementation of MSE-optimal bandwidth. Default choice.
- mse-rot** ROT implementation of MSE-optimal bandwidth.
- imse-dpi** second-generation DPI implementation of IMSE-optimal bandwidth.
- imse-rot** ROT implementation of IMSE-optimal bandwidth.
- ce-dpi** second generation DPI implementation of CE-optimal bandwidth.
- ce-rot** ROT implementation of CE-optimal bandwidth.

Note: MSE = Mean Square Error; IMSE = Integrated Mean Squared Error; CE = Coverage Error; DPI = Direct Plug-in; ROT = Rule-of-Thumb.
 Default is **bwselect(mse-dpi)**. For details on implementation see [Calonico, Cattaneo and Farrell \(2019\)](#).

bwcheck(#) specifies an optional positive integer so that the selected bandwidth is enlarged to have at least # effective observations available for each evaluation point.

imsegrid(#) number of evaluations points used to compute the IMSE bandwidth selector. Default is 30 points.

vce(*vcetype* [*vceopt1*]) specifies the procedure used to compute the variance-covariance matrix estimator. Options are:

- vce**(*nn* [*nnmatch*]) for heteroskedasticity-robust nearest neighbor variance estimator with *nnmatch* indicating the minimum number of neighbors to be used.
- vce**(*hc0*) for heteroskedasticity-robust plug-in residuals variance estimator without weights.
- vce**(*hc1*) for heteroskedasticity-robust plug-in residuals variance estimator with *hc1* weights.
- vce**(*hc2*) for heteroskedasticity-robust plug-in residuals variance estimator with *hc2* weights.
- vce**(*hc3*) for heteroskedasticity-robust plug-in residuals variance estimator with *hc3* weights.
- vce**(**nncluster** *clustervar* [*nnmatch*]) for cluster-robust nearest neighbor variance estimation using with *clustervar* indicating the cluster ID variable and *nnmatch* matches indicating the minimum number of neighbors to be used.
- vce**(**cluster** *clustervar*) for cluster-robust plug-in residuals variance estimation with degrees-of-freedom weights and *clustervar* indicating the cluster ID variable.

Default is **vce(nn 3)**.

level(#) specifies confidence level for confidence intervals. Default is **level(95)**.

bwregul(#) specifies scaling factor for the regularization term added to the denominator of the bandwidth selectors. Setting **bwregul(0)** removes the regularization term from the bandwidth selectors. Default is **bwregul(1)**.

separator(#) draws separator line after every # variables; default is **separator(5)**.

interior option to set all evaluation points to be interior points. This option affects only data-driven bandwidth selection via [lpbwselect](#).

covgrid option to compute two covariance matrices (*cov_us* and *cov_rb*) for classical and robust covariances across point estimators over the grid of evaluation points.)

plot generates the local polynomial regression plot.

`genvars` generates new variables storing the following results.

- `lprobust_eval` evaluation points.
- `lprobust_h` bandwidth h .
- `lprobust_b` bandwidth b .
- `lprobust_nh` effective sample size.
- `lprobust_gx_us` conventional local polynomial estimate.
- `lprobust_se_us` conventional standard error for the local polynomial estimator.
- `lprobust_gx_bc` bias-corrected local polynomial regression estimate.
- `lprobust_se_rb` robust standard error for the local polynomial estimator.
- `lprobust_ci_l_rb` lower end value of the robust confidence interval.
- `lprobust_ci_r_rb` upper end value of the robust confidence interval.

`graph_options` (*gphopts*) specifies graphical options to be passed on to the underlying graph command.

Setup

`. webuse motorcycle`

Local linear regression with second-generation DPI implementation of MSE-optimal bandwidth

`. lprobust accel time`

Same as above, but generating a plot and the corresponding output variables

`. lprobust accel time, plot genvars`

Saved results

`lprobust` saves the following in `e()`:

Scalars

<code>e(N)</code>	original number of observations
<code>e(p)</code>	order of the polynomial used for estimation of the regression function

Macros

<code>e(varname)</code>	name of variable
<code>e(clustvar)</code>	name of cluster variable
<code>e(bwselect)</code>	bandwidth selection choice
<code>e(kernel)</code>	kernel choice
<code>e(vce)</code>	vce choice

Matrices

<code>e(Result)</code>	estimation result
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References

- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2018. [On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference](#). *Journal of the American Statistical Association*, 113(522): 767-779.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2019. [nprobust: Nonparametric Kernel-Based Estimation and Robust Bias-Corrected Inference](#). *Journal of Statistical Software*, 91(8): 1-33. doi: 10.18637/jss.v091.i08.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2022. [Coverage Error Optimal Confidence Intervals for Local Polynomial Regression, Bernoulli](#), forthcoming.
- Fan, J., and Gijbels, I. 1996. *Local Polynomial Modelling and Its Applications*, London: Chapman and Hall.
- Wand, M., and Jones, M. 1995. *Kernel Smoothing*, Florida: Chapman & Hall/CRC.

Authors

Sebastian Calonico, University of California, Davis, CA. scalonico@ucdavis.edu.

Matias D. Cattaneo, Princeton University, Princeton, NJ. cattaneo@princeton.edu.

Max H. Farrell, University of California, Santa Barbara, CA.
maxhfarrell@ucsb.edu.